

SAI ABHINAV

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Portfolio - saixabhinav.vercel.app | Github - github.com/SaixAbhinav | LinkedIn - linkedin.com/in/sai-abhinav

SUMMARY

I'm a MCA student building and deploying end-to-end ML systems across reinforcement learning, CNNs, and LLM workflows. Deployed a live RL traffic-signal optimizer (PPO + SUMO) that cut average wait time by up to 91% vs. a fixed-timer baseline and beat max-pressure control by ~19% under asymmetric demand. Seeking a Junior AI/ML Engineer role to ship production ML and GenAI systems.

PROJECTS

SmartSignal — RL Traffic-Signal Optimization | [Demo](#) | [GitHub](#)

Python · PPO (Stable-Baselines3) · SUMO · FastAPI · Render

- Trained a PPO agent to convergence (500K timesteps, ~17 min on a CPU laptop, 8 parallel SUMO envs) that cut wait time ~91% vs. a fixed-timer baseline and beat max-pressure control by ~19% on asymmetric rush-hour demand, validated across 3 traffic profiles over 5 random seeds.
- Extended to a 4-junction shared-policy grid (2,880 decision points/episode) that outperformed max-pressure by 22–35% and produced emergent green-wave coordination with no hand-coded offsets, trained in ~6 minutes.
- Shipped a FastAPI + WebSocket dashboard running all controllers side-by-side on identical seeded traffic in real time, deployed live on Render.

Workflow Copilot — Local-First Document & Inbox Automation | [GitHub](#)

Python · Ollama (local LLM) · PyQt5 · Gmail & Calendar API (OAuth2) · SQLite

- Built an Auto-triage mode that scans up to 15 emails per window (4/12/24/48h), truncating each to 150 words before a parallel LLM call, then ranks results by urgency and pushes prioritized tasks to Google Calendar.
- Designed a multi-account Gmail/Calendar integration with per-account OAuth token isolation, supporting an arbitrary number of linked accounts.
- Implemented structured-output prompting across 4 workflows (Summary, Tasks, Insights, Compare) with a 300-word sentence-boundary-aware chunker for long documents; ~3,000 lines of application code covered by 19 passing tests.

EXPERIENCE

Data Analyst Intern — IBM SkillsBuild | [GitHub](#)

Jun 2024 – Aug 2024

- Built and tuned a CNN for skin-cancer image classification, reaching 94% accuracy through preprocessing, augmentation, and hyperparameter tuning.
- Worked in a 6-person team and presented model results to mentors and stakeholders.

EDUCATION

Master of Computer Applications (MCA)

2025 – 2027

Vivekananda Institute of Professional Studies, Delhi · CGPA 8.6

Bachelor of Computer Applications (BCA)

2022 – 2025

Vivekananda Institute of Professional Studies, Delhi · CGPA 7.6

TECHNICAL SKILLS

Languages: Python, SQL

ML / Deep Learning: PyTorch, TensorFlow, scikit-learn; CNNs / Computer Vision; Feature Engineering; Model Optimization

Reinforcement Learning: PPO, Stable-Baselines3, SUMO

GenAI / LLM: Ollama (Local LLMs), Prompt Engineering, RAG Fundamentals

Deployment & Tools: Git / GitHub, FastAPI, REST APIs, Render, PyQt5

Data: Pandas, NumPy, Matplotlib